

CESC 410L — Lab 1: Artifact Collection

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Section 01

Fall 2026

This document collects the artifacts the three Lab 1 tasks call for, in the order the handout lists them.

Project Management Task

Artifact 1 — output of `dsp26 --help`

```
Usage: dsp26 [OPTIONS] COMMAND [ARGS]...

Commands
  plot-sinusoids-td-fd  Plot time-domain waveform and frequency-domain
                        spectrum of sinusoids.
  play-plot-audio       Play and plot audio signal.
                        Use --path-name to load audio file from.
```

Artifact 2 — the Audio Player window



Figure 1: The Audio Player window shown while the audio file plays.

Properties reported for the supplied file

The command prints the WAV header fields before drawing anything:

```
Number of channels: 1
Sample width (bytes): 2
Frame rate (sample rate): 16000
Number of frames: 160000
Duration (seconds): 10.0
```

One channel of 16-bit samples at 16 kHz, and $160000/16000 = 10.0$ s. The content is two constant-amplitude sinusoids at 440 Hz and 554 Hz, which is what the five figures below show.

Artifacts 3–7 — the five figures

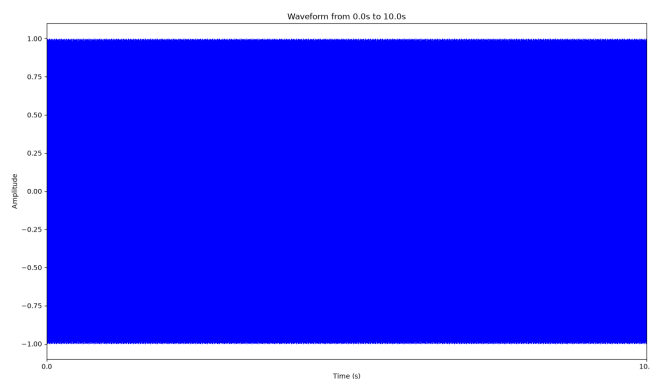


Figure 2: Handout Figure 1 — waveform of the whole file, 0 to 10 s. It fills as a solid block between ± 1.0 because the signal is a sustained tone of constant amplitude: there is no onset, no decay and no silence anywhere in the file for the envelope to show.

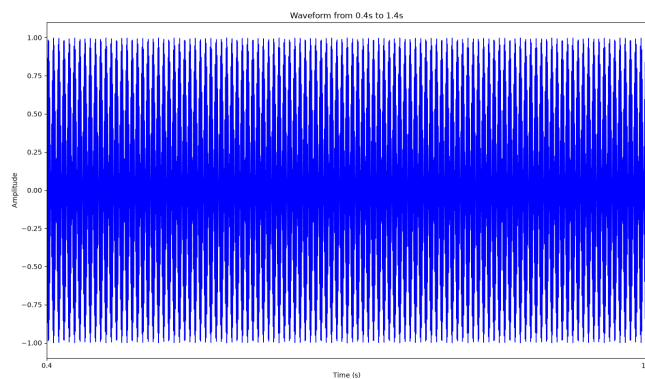


Figure 3: Handout Figure 2 — waveform zoomed to 0.4–1.4 s. The zoom resolves what Figure 1 could not: the two tones interfere, and their sum swells and collapses at their difference frequency, $554.4 - 440.0 = 114.4$ Hz. The window is one second wide and contains 114 such lobes.

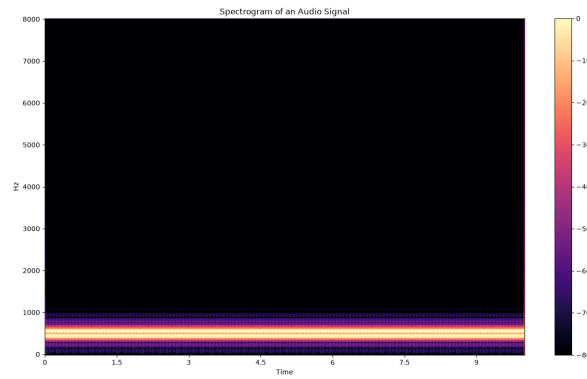


Figure 4: Handout Figure 3 — spectrogram, linear frequency axis. All of the energy lies in one band near the bottom of the 0–8 kHz range and the plot is empty above roughly 1 kHz. On this axis the two tones are 114 Hz apart out of 8000 and cannot be told apart.

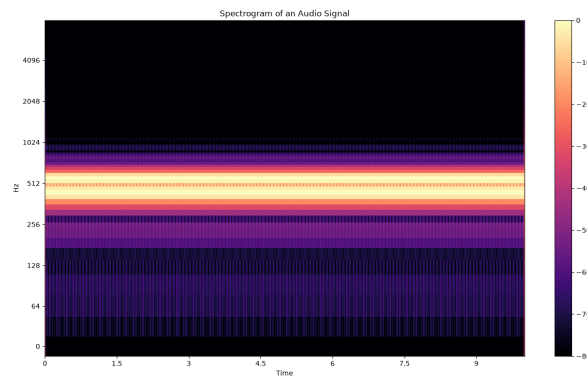


Figure 5: Handout Figure 4 — spectrogram, logarithmic frequency axis. The same data as Figure 3, but the expanded low end separates the band into the two distinct horizontal lines it is made of, either side of the 512 Hz tick. Both run flat for the full ten seconds, with no harmonics above them.

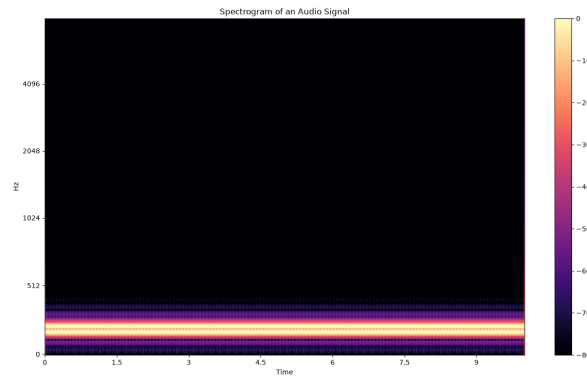


Figure 6: Handout Figure 5 — spectrogram, mel frequency axis. The mel scale is near-linear below 1 kHz, so the two tones stay close together and the plot is black above about 512 Hz — the warping that helps with speech has little to work with on a signal this sparse.

Experimental Task

A ten-second excerpt of music, converted to a single channel at a 16 kHz sample rate. The properties reported by the application confirm the format:

```
Number of channels: 1
Sample width (bytes): 2
Frame rate (sample rate): 16000
Number of frames: 160000
Duration (seconds): 10.0
```

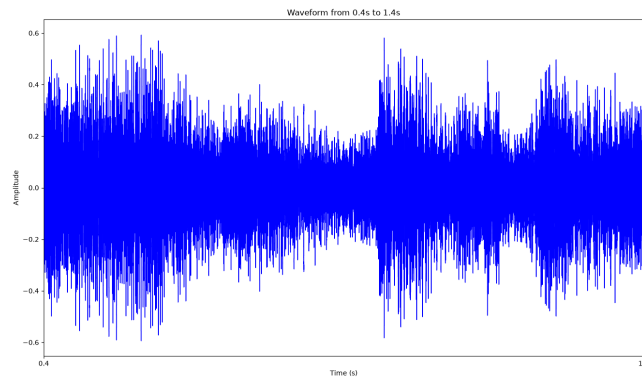


Figure 7: The second (zoomed-in) waveform figure, for the music file.

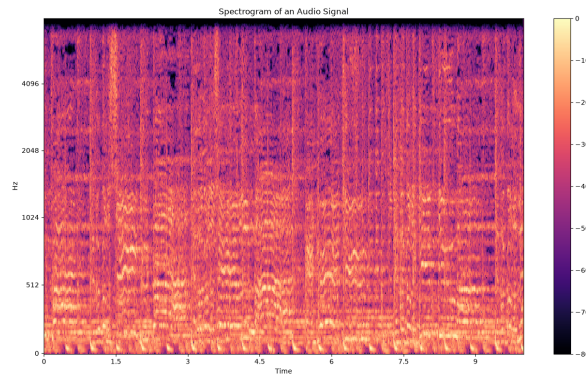


Figure 8: The last spectrum figure, for the music file.

Programming Task

Code snippet

```
start_s: float = typer.Option(
    0.0, "--start-s",
    help="Start time in seconds for the zoomed waveform figure."),
end_s: float = typer.Option(
    None, "--end-s",
    help="End time in seconds for the zoomed waveform figure."),
```

Listing 1: `app_cli.py` — the two new options

```
if start_s == 0.0 and end_s is None:
    zoom_start, zoom_end = HANDOUT_ZOOM_S      # (0.4, 1.4)
else:
    zoom_start, zoom_end = start_s, end_s

fig2 = plot_audio_sig_in_td(audio, sample_rate, zoom_start, zoom_end)
```

Listing 2: `recipes.py` — passing them through

Code explanation

Each `typer.Option` binds a Python parameter to a command-line flag. The first argument is the value used when the flag is absent, the second is the flag’s spelling, and the annotated type drives both conversion and validation — so `--start-s 0.5` arrives as the float `0.5`, and a non-numeric value is rejected before any code runs. The help strings become the text shown by `--help`.

The defaults are those of `plot_audio_sig_in_td` itself, as the task requires: zero, and “to the end of the file”. Since that pair means the whole file, supplying neither would make the zoomed figure a duplicate of the full-waveform figure, so when neither is given the handout’s own 0.4–1.4s window is used instead; any explicit window is passed through unchanged.

No plotting code was modified. `plot_audio_sig_in_td` already accepted an arbitrary window, so the task was to expose that capability at the command line, not to implement it.

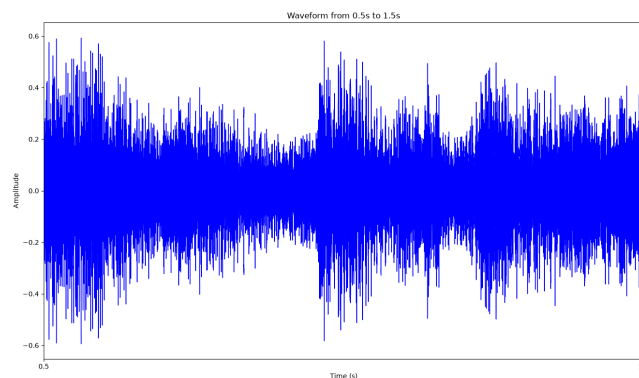


Figure 9: The music file, zoomed from 0.5 to 1.5 seconds, produced with `--start-s 0.5 --end-s 1.5`.